

Solve the room, don't template it.

A constraint solver — not a set of fixed layouts — reads your room, your style, your budget and, optionally, an inspiration photo, then computes a floor plan that is physically valid before it is ever drawn.

1



Describe the room

Length, width, ceiling, door wall,
facing direction

2



Set the brief

Style, must-have fixtures, budget
ceiling, Vastu on / off

3



Solve, don't guess

Every wall × every fixture size ×
every door swing is scored for
clearance, circulation and cost

4



Draw from one truth

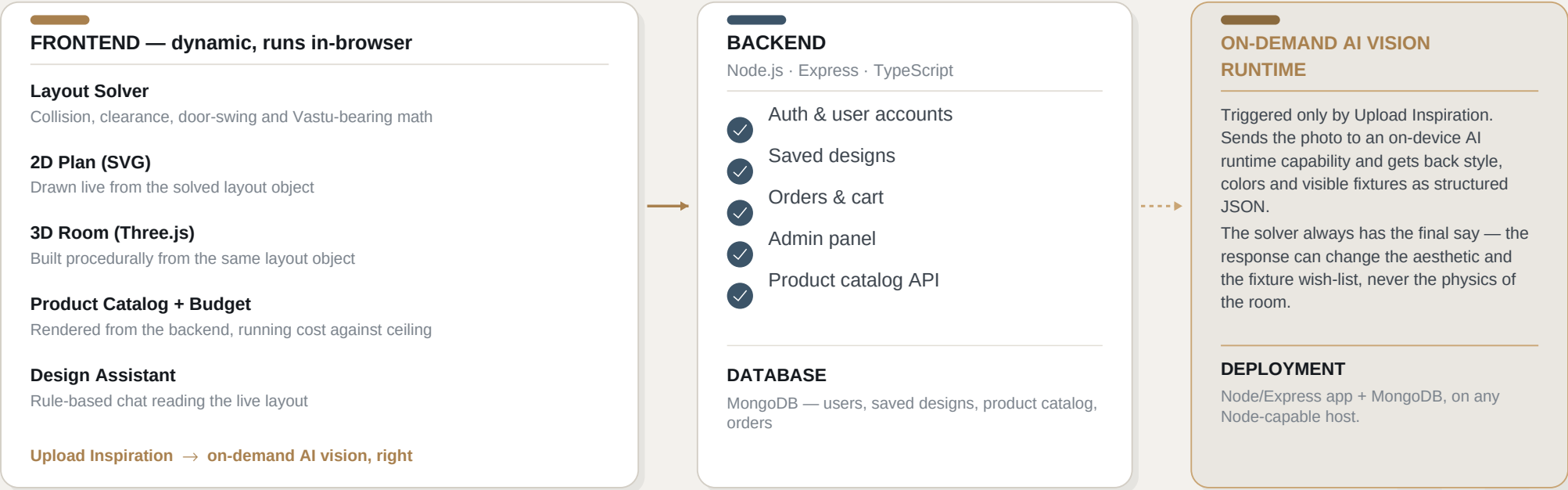
The winning layout renders as SVG
(2D) and Three.js (3D) — never out
of sync, because both read the same
object

WHAT COMES OUT

- ✓ Collision-free 2D floor plan
- ✓ Itemized KOHLER product list + total cost
- ✓ Plain-language layout report — what was adjusted and why
- ✓ Matching 3D room, same data
- ✓ Vastu compliance read-out

A full-stack build; the physics stays client-side.

Accounts, saved designs, orders and the admin panel are handled by a real backend. The layout math — collision, clearance, door-swing and Vastu-bearing scoring — still runs entirely in the browser, so it is never at the mercy of a network round-trip.



Small where it can be, real where it has to be.

A deliberately light frontend for the parts that must stay instant and offline-safe, and a proper backend for the parts that need to persist.

FRONTEND

- **Vanilla JavaScript**

App logic, solver, state — zero framework

- **SVG**

2D floor plan, generated live

- **Three.js (CDN)**

Procedural 3D room + materials

- **Hand-rolled CSS**

Design tokens, light/dark theme

BACKEND & DATA

- **Node.js + Express + TypeScript**

REST API — accounts, designs, orders, admin

- **MongoDB**

Users, saved designs, product catalog, orders

- **Token-based auth**

Protects account, order and admin routes

- **Any Node-capable host**

Render / Railway / EC2 / on-prem

INTELLIGENCE

- **Rule-based solver**

Deterministic layout scoring — not an LLM call

- **Regex-driven chat**

Design assistant reads the live layout object

- **On-device AI vision runtime**

Real image analysis, Upload Inspiration only

- **Dynamic product catalog**

KOHLER SKUs, prices, categories — served from MongoDB

A bathroom you can trust before it's built.

Most “AI design” tools pick from a gallery of pre-made looks. This one computes the room from first principles — so every layout it hands back already clears the door swing, the walk, and the wall it's on.

1

Solved, not styled

Fixtures aren't dragged onto a template — clearances, circulation and door access are checked before a layout is ever shown.

2

A photo becomes a brief

Upload Inspiration turns a real bathroom photo into structured style + fixture data via an on-device AI vision runtime, then re-solves your actual room around it — never just an overlay.

3

Honest about what's dropped

When something from the brief won't fit, the layout report says exactly why and what's carrying that function instead — no silent compromises.

4

Vastu as a real constraint

Direction-aware scoring, not a decorative label — toggle it and the solver re-optimizes around the bearings.

5

2D and 3D from one object

The floor plan and the room are two views of the same solved layout, so they can never drift out of sync.

6

Persistence without losing the guarantee

Accounts, saved designs and orders live safely behind the Express API in MongoDB — but the safety-critical clearance math never leaves the browser, so the network can't get it wrong.